

Chun-Yuan Chen

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Education

National Sun Yat-Sen University

Master of Science in Electrical Engineering

Kaohsiung, Taiwan

09/2023 – Present (Expected 07/2026)

- **Course Highlights:** Design and Analysis of Algorithms, Wireless and Mobile Networks

- **Exchange Student at Kyushu University**

Fukuoka, Japan

– *Graduate School of Information Science and Electrical Engineering* *09/2025 – 03/2026*

- * **Course Highlights:** Numerical Analysis for Information Science, Machine Learning

National Changhua University of Education

Bachelor of Business Administration in Information Management

Changhua, Taiwan

02/2020 – 07/2022

- **Course Highlights:** Big Data Analysis in Practice, Databases

National Pingtung University

Bachelor of Engineering in Computer and Communication

Pingtung, Taiwan

09/2018 – 01/2020

(Transferred to NCUE to complete degree)

Master Thesis

Multi-Criteria Handover in LEO Satellite Networks via CTDE-Based Multi-Agent Reinforcement Learning

Proposed a handover strategy for LEO satellite networks using multi-agent reinforcement learning to optimize network performance and reduce handover frequency.

- Utilized Centralized Training with Decentralized Execution framework to train agents representing ground terminals for making their own handover decisions
- Designed unique reward function considering factors like QoS, load balancing, and handover costs
- Evaluated performance through simulations, demonstrating improved network throughput and reduced handover frequency compared to traditional methods

Projects

Feature Selection for Yield Prediction in Semiconductor Manufacturing

Kyushu University – Machine Learning Final Project (01/2026)

- Designed and executed a reproducible ML pipeline to address severe class imbalance on the industrial SECOM dataset using Scikit-learn.
- Compared filter-based and embedded feature selection techniques under a unified framework, utilizing Balanced Error Rate (BER) to maximize prediction accuracy.

- Presented findings to faculty at the Kyushu University Graduate School of Information Science and Electrical Engineering.

Stabilized Deep Reinforcement Learning via Double-DQN (DDQN)

Utilized Python and PyTorch to implement and compare standard DQN and Double DQN architectures for stable value estimation.

- Implemented a Double Deep Q-Network (DDQN) architecture from scratch to mitigate the overestimation bias inherent in standard DQN algorithms.
- Integrated a Convolutional Neural Network (CNN) feature extractor with experience replay and decoupled target networks to stabilize training trajectories.
- Evaluated and validated algorithm convergence using the high-dimensional, real-time physics environment of Flappy Bird, achieving robust high-score baselines.

Network Packet Transfer via TCP and UDP

Utilized C++ to implement TCP and UDP protocols for reliable and fast data transfer over a network.

- Implemented TCP features such as connection establishment, data transfer, and connection termination
- Implemented UDP features for fast, connectionless data transfer without reliability guarantees
- Evaluated performance of both protocols under different network conditions and data sizes

Experience

Teaching Assistant

Design and Analysis of Algorithms,

Kaohsiung, Taiwan

02/2025 – 07/2025

- Managed classroom logistics, technical equipment setup, and environmental controls to ensure optimal conditions for graduate/undergraduate courses.
- Assisted course instructors with administrative tasks, attendance tracking, and materials distribution.
- Served as a reliable point of contact between students and faculty to facilitate smooth daily operations.

Skills

Programming Languages: C++, Python

Tools & Technologies: PyTorch, Scikit-learn, Git, Docker, AWS

Languages: Mandarin (Native), English (Fluent; TOEIC Gold Certificate), Japanese (Basic)